

This public disclosure focuses on the critical role of **Chiral Cellulose Nanocrystals (CNCs)** in the self-assembly of the magnonic-electric vortex coating, ensuring this specific geometric mechanism is established as prior art.

The Chiral "Screw" (Self-Assembly Mechanism)

The use of **Cellulose Nanocrystals (CNCs)** is the primary method for forcing the Magnonic Ink into a helical geometry during the drying process.

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- **Steric Hindrance:** CNCs possess a natural helix that, as the ink carrier (ethanol/DMF) evaporates, utilizes **steric hindrance** to force the surrounding Perovskite-Magnon crystals to align along this spiral path.

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- **The "Automagic" Helical Assembly:** This is defined as an "Ambient Temperature Assembly" process where the internal energy of the CNC molecules builds the structure without external energy refinement.

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- **Geometric Locking:** As the solvent evaporates, the CNCs cause the entire triple-layer suspension to lock into a permanent macroscopic helical "Screw" structure.

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Defensive Prior Art Establishment

To prevent "weasels" from claiming novelty on the geometry of the system, this disclosure explicitly states:

- **Inherent Property:** The "spinning" or spiral geometry is an **inherent property** of the natural chemistry when CNCs are combined with the perovskite-magnon core.

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- **Impedance Matching:** The specific helical pitch created by the CNCs is designed to provide a physical **impedance match** for the 137.036 vorticity of light, allowing the capture of spin-wave torque.

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- **Blocking Derivative Patents:** By documenting the use of CNCs to orient magnonic dopants in a helical pattern, any future attempts to patent "chiral thermoelectric coatings" or specific "pitch lengths" of such spirals are rendered obvious and unpatentable under 35 U.S.C. § 103.

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Synthesis Integration (Solution B)

In the synthesis process, CNCs are the foundational "Aligner":

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1. **Preparation:** YIG nanoparticles and Graphene are sonicated directly into the **Chiral Aligner (CNCs)** to set the resonant frequency.

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2. **Coherent Bridge:** This aligner ensures that once the ink is applied in a vortex spray, the magnons have a coherent "bridge" to travel through, facilitating the Spin Seebeck Effect and its conversion to electrical power.

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